

5. Project Development Phase

1. INTRODUCTION

The Project Development Phase is the fifth stage in the development of **LegalEase AI**. In this phase, the ideas, requirements, design, and plans created during the previous stages are converted into an actual working application.

The project is developed using Visual Studio Code (VS Code) as the main development environment. The frontend is created for user interaction, while the backend is implemented to receive and process user requests. AI/LLM functionality is then integrated into the application to process general legal queries and provide simplified information.

The required dependencies and configuration files are also added during this phase to connect and integrate the different components of the application.

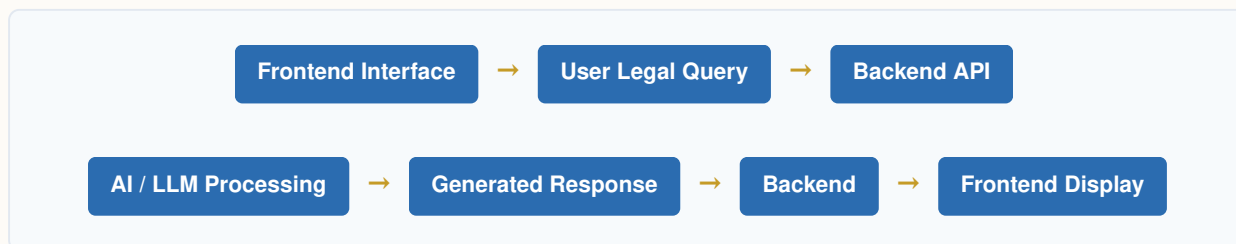
2. OBJECTIVES

The main objectives of the Project Development Phase are:

- Develop the LegalEase AI application using VS Code.
- Create the frontend user interface.
- Implement backend APIs.
- Integrate AI/LLM functionality.
- Connect user legal queries with the AI processing system.
- Add required dependencies and configuration files.
- Integrate the different application modules.

3. DEVELOPMENT WORKFLOW

The core workflow routes user interactions through backend validation, AI interpretation, and response formatting:



4. DETAILED DEVELOPMENT

4.1 Frontend Development

The frontend is developed to provide a simple interface for users. It allows users to enter their general legal questions and view the response generated by the AI system.

A basic HTML interface is constructed as follows:

```
<!DOCTYPE html>
<html>
<head>
  <title>LegalEase AI</title>
</head>
<body>
  <h1>LegalEase AI</h1>
  <textarea id="question" placeholder="Enter your legal question"></textarea>
  <button onclick="askQuestion()">Ask LegalEase AI</button>
  <p id="response"></p>
</body>
</html>
```

4.2 Backend Development

The backend is implemented to receive requests from the frontend and process them. A basic Flask API endpoint is established:

```
from flask import Flask, request, jsonify

app = Flask(__name__)

@app.route("/ask", methods=["POST"])
def ask_question():
    data = request.get_json()
    question = data.get("question")

    if not question:
        return jsonify({"error": "Question is required"}), 400

    return jsonify({
        "question": question,
        "response": "AI response will be generated here."
    })

if __name__ == "__main__":
    app.run(debug=True)
```

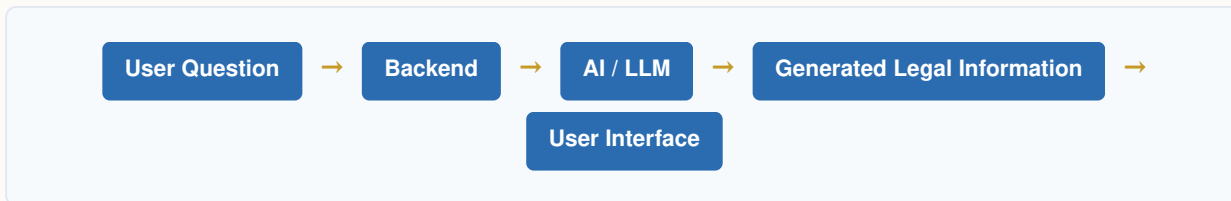
Code Explanation:

- Flask initializes the backend server application.
- The /ask route receives POST payload containing the user's legal query.
- `request.get_json()` parses incoming client requests.

- Input validation checks ensure non-empty queries.
- The JSON response structure prepares for integrated AI results.

4.3 AI/LLM Integration

The AI/LLM module is integrated into the backend so that user questions can be processed automatically:



The AI component is directly connected to the query-processing system, ensuring natural language processing capabilities for legal text simplification.

4.4 Dependencies and Configuration

Dependencies are declared in `requirements.txt`:

```
Flask
python-dotenv
requests
```

Environment configurations are stored securely inside `.env`:

```
AI_API_KEY=your_api_key_here
```

5. EXPECTED OUTCOME

At the conclusion of the Project Development Phase, all functional components are integrated into a cohesive application featuring a responsive user interface, backend routes, operational LLM API execution, dependency resolution, and secure key configurations.

6. PHASE CONCLUSION

The Project Development Phase converts the planned design of LegalEase AI into a working application. Connecting the frontend, backend, and AI/LLM components completes the operational foundation, readying the application for the **Project Testing Phase**.